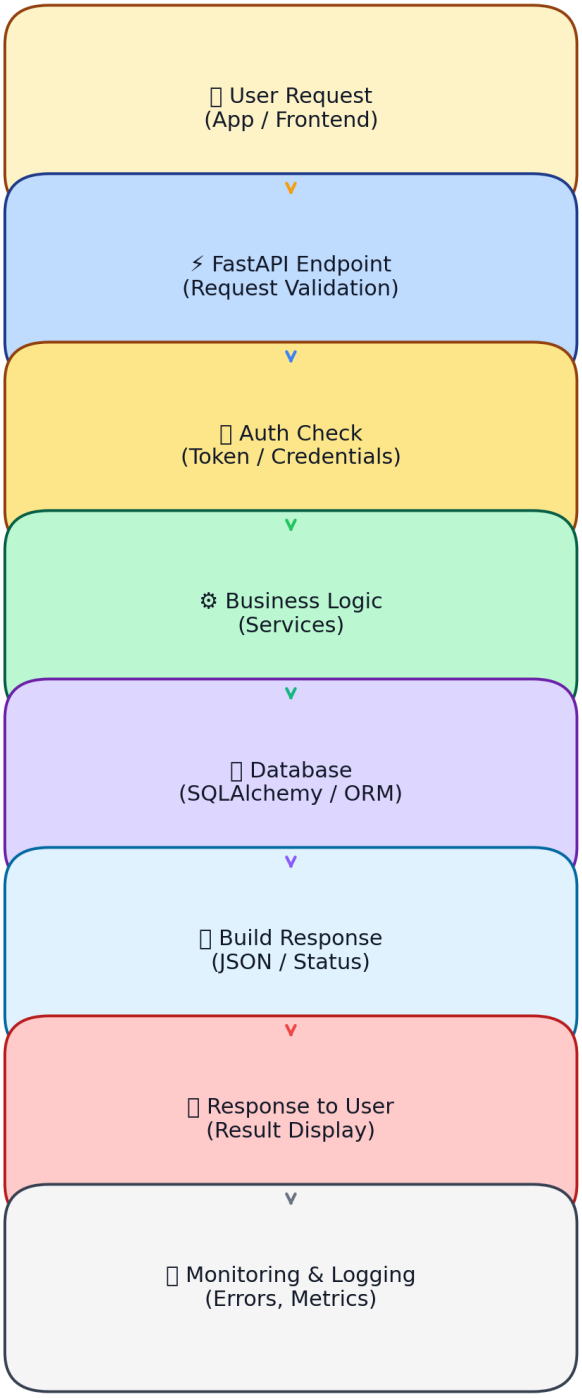
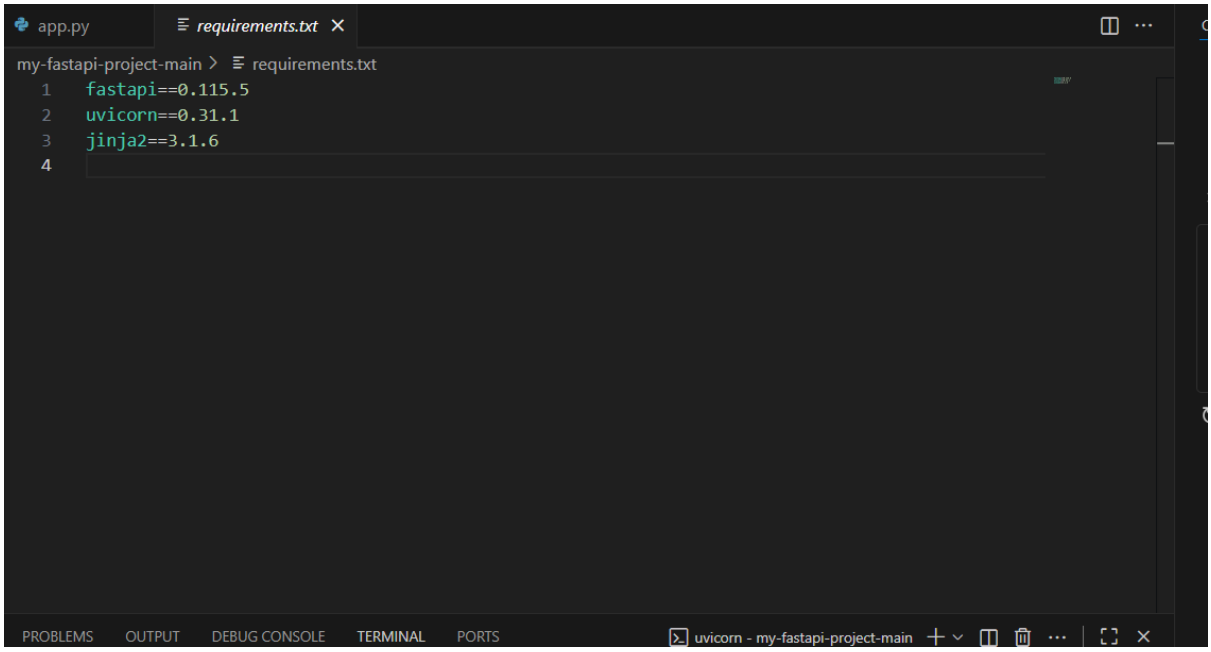
USER WORKFLOW (FASTAPI PROJECT)

1. **User Request**
   - A user opens the frontend (React / HTML / mobile app) and sends a request.
   - Example: *User submits login credentials or form data.*
2. **API Gateway (FastAPI Endpoint)**
   - The request hits a FastAPI route (@app.post("/login") or similar).
   - Request validation via **Pydantic schemas**.
3. **Decision: Authentication**
   - If valid → continue.
   - If invalid → return **error response** (401 Unauthorized).
4. **Business Logic (Processing Layer)**
   - FastAPI calls service functions (controllers).
   - Executes business rules (e.g., create user, predict result, store record).
5. **Database Interaction**
   - Query/insert/update via ORM (e.g., SQLAlchemy).
   - Decision: *Does record exist?* → Yes → update, No → create new.
6. **Response Building**
   - Data converted into schema → JSON response.
   - Include status codes (200 OK, 400 Bad Request, etc.).
7. **User Output**
   - Frontend displays success/failure messages.
   - Example: *"Login successful"* or *"Invalid password"*.
8. **Monitoring & Logging**
   - Requests & responses logged.
   - Errors stored for debugging.

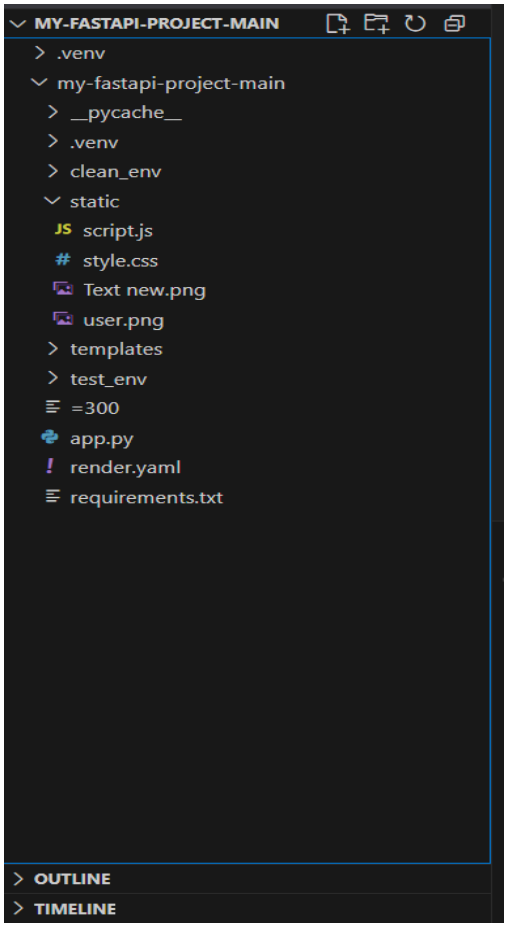
FastAPI Project – User Workflow (Vertical Infographic)

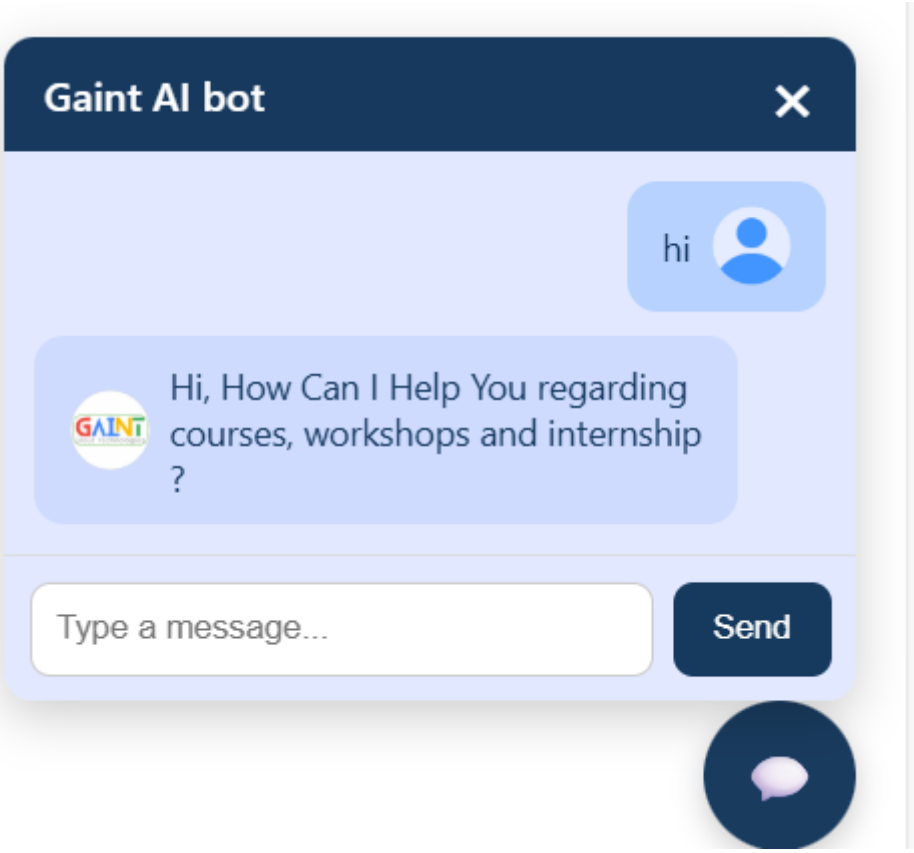


Installations:



Project structure



Outputs:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS        uvicorn - my-fastapi-project-main
PS E:\Gaint Works\internship projects\my-fastapi-project-main> cd .\my-fastapi-project-main\
PS E:\Gaint Works\internship projects\my-fastapi-project-main\my-fastapi-project-main> cd .\.venv\
PS E:\Gaint Works\internship projects\my-fastapi-project-main\my-fastapi-project-main\.venv> cd .\Scripts\
PS E:\Gaint Works\internship projects\my-fastapi-project-main\my-fastapi-project-main\.venv\Scripts> ./Activate
(.venv) PS E:\Gaint Works\internship projects\my-fastapi-project-main\my-fastapi-project-main\.venv\Scripts> cd ..
(.venv) PS E:\Gaint Works\internship projects\my-fastapi-project-main\my-fastapi-project-main\.venv> cd ..
(.venv) PS E:\Gaint Works\internship projects\my-fastapi-project-main\my-fastapi-project-main> uvicorn app.main:app --
reload
INFO:     Will watch for changes in these directories: ['E:\\Gaint Works\\internship projects\\my-fastapi-project-main
\\my-fastapi-project-main']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [17516] using WatchFiles
```